MIL-S-81749(AS)
21 November 1969
Superseding
XWS-4948
6 April 1964

MILITARY SPECIFICATION

SPRAY OUTFIT, PAINT, AIRLESS, HYDRAULIC PRESSURE, PORTABLE

This specification has been approved by the Naval
Air Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one type of portable, hydraulic pressure, airless paint spray outfit and accessories for spraying of unheated coating materials such as enamel, lacquer, oil base paints, vinyl base paints, dope, etc.

2. APPLICABLE DOCUMENTS

2.1 Government documents normally furnished - The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-E-529	Enamel, Alkyd, Semi-Gloss
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-621	Box, Wood, Nailed and Lock-Corner
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000/1	Drawings, Engineering and Associated Data
MIL-A-8421	Air Transportability Requirements, General Specification For

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specification and Standard, Order of Precedence for the Selection of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing with Appropriate Test Methods
MS33586	Metal, Definition of Dissimilar

PUBLICATIONS

National Bureau of Standards

Handbook H-28	Screw-Thread Standards for Federal Services

(When requesting applicable documents, refer to both title and number. Copies of unclassified documents may be obtained from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120. Requests for copies of classified documents should be addressed to the Naval Publications and Forms Center, via the cognizant Government representative.)

3. REQUIREMENTS

3.1 First article - The spray outfit furnished under this specification shall be a product which has been inspected and passed the first article inspections specified herein (see 4.3).

3.2 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials - Materials shall conform to applicable specifications and shall be as specified herein. Materials which are not covered by applicable specifications shall be of the best quality and entirely suitable for the purpose intended.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.3.1 Metals - All metal parts shall be of corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion by fuels, oils, salt spray, water, or atmospheric conditions.

3.3.1.1 Dissimilar metals - Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.3.2 Fungus-proof material - Materials that are nutrients of fungi shall not be used.

3.4 Design and construction - The spray outfit shall be designed and constructed in such a manner as to provide a compact, self-contained, portable, convenient, and safe means of spraying enamel, lacquer, oil base paints, vinyl base paints, dope, and other coating materials to surfaces. The spray outfit shall be capable of drawing and spraying paint from original standard one (1), five (5), and ten (10) gallon containers. The spray outfit shall be complete and shall require only an air supply with an adequate paint supply to completely perform its intended function. The spray outfit shall be rugged, of substantial design, and shall be capable of continuous operation without excessive wear, fatigue, maintenance, and repair.

3.4.1 Safety devices - The equipment shall be fitted with suitable safety devices to prevent injury to operating personnel. The equipment shall use electrically conductive hoses (air and fluid) to maintain electrical continuity between spray gun, pumping unit, air hose, and air hose inlet fitting to ground any static charges which may be generated by the spray gun. A safety trigger lock shall be provided as described in 3.8.4.4.

3.4.2 Fastening devices - All screws, pins, bolts, and similar parts shall be designed with adequate means to prevent loss of proper torque and adjustment. These parts shall not work loose due to vibration. All such parts, when subject to removal or adjustment, shall not be swaged, peened, staked, or otherwise permanently deformed.

3.4.2.1 Screw threads - All screw threads shall have a smooth finish without imperfections such as burrs or projections. All screw threads and parts of the gun and equipment which are frequently assembled and disassembled in normal usage shall be right-hand. The assembly and disassembly of hoses and spray gun shall be accomplished without the use of special tools or wrenches. Threads shall be in accordance with applicable requirements of Handbook H-28. Threads shall be NPSM, type I and NPT sizes.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the current version before use.

3.4.3 Welding, brazing, soldering, peening, or plugging - When such operations are required in the assembly of an item, the welding, brazing, soldering, peening, or plugging shall be performed in accordance with the best recognized commercial practice. In no event shall welding, brazing, soldering, peening, or plugging be resorted to as a repair measure. Any welded parts subject to the high pressure in the coating material line shall be stress relieved.

3.4.4 Castings - All castings shall be of uniform quality, adequately braced, trussed, or otherwise reinforced and completely free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Castings shall be free of flashings and sprues and shall be thoroughly cleaned of all scale, molding sand, and other extraneous materials. All exterior surface of castings shall be sanded or ground to provide a smooth, blended finish surface. All corners shall be chamfered, beveled, rounded, or otherwise blended to eliminate unnecessary service hazards to the operator.

3.4.4.1 Seasoning - All castings, after being cast, shall be aged either naturally or artificially to insure dimensional stability.

3.4.5 Forgings - Forgings shall be free from scale, inclusions, mismatching, and other defects which may affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings and forgings shall be adequate to meet the performance requirements specified herein.

3.5 Maintainability - All working parts and components shall be designed to afford maximum operational performance with minimum maintenance. Adjustment shall be readily accomplished by operating personnel without special training, and without the need of special tools. All parts subject to wear, breakage, or loss of adjustment shall be readily accessible for adjustment, repair, or replacement.

3.6 Air transportability - The spray outfit shall be designed for air transportability in accordance with MIL-A-8421.

3.7 Performance characteristics - The spray outfit shall meet the performance requirements specified in Section 4 when subjected to the applicable tests.

3.8 Details of components - The spray outfit shall consist of a cover assembly and paint container, pump, paint filter, spray gun, hoses, air filter and oiler, carriage assembly, and tool box. The pump with air filter and oiler, and hose shall be mounted on the cover assembly.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.8.1 Cover assembly and paint container - The cover assembly shall be designed to fit and hold securely on a standard 10 gallon paint container. One paint container with two handles shall be provided with each spray outfit. Means shall be provided to support the cover assembly in a raised position for cleaning. The cover assembly shall have a closeable inspection port for visual checking of mixing operations and for the addition of spray material, as required. The closeable inspection port cover shall be secured to the tank cover in such a manner that it cannot be accidentally removed, and will seal the port so that spraying fluids cannot pass the seal when the unit is operating. The seal for the cover assembly shall be made of a rubber compound or any other suitable material having resistance to the effects of solvents, paints, thinners, and various other types of spray fluids.

3.8.2 Pump - The hydraulic pump shall be air driven, one or two piston type, with a stroke that assures constant spray pattern. The pump shall be of the double acting immersion type and shall be self-contained. All components subject to the increased pressures shall be constructed of stainless steel or other suitable heavy-plated materials, for resistance to the high pressures and chemical reaction from the coating materials, thinners, or solvents. The pump shall be designed for tank bottom suction and shall be provided with a 100-mesh inlet filter screen. Additional inlet filter screens in the desired mesh shall be ordered separately, as required by the procuring activity (see 6.2). The pump shall be capable of providing a six foot head while maintaining its maximum rated capacity.

3.8.2.1 Control valve - An air control valve or other means shall be provided for "on" and "off" control of the pump. The valve shall be leak proof so that the pump will not function while the operator is filling the paint container or adding solvents.

3.8.2.2 Pump check valves - The check valve seats and the check valve balls shall be contructed of a suitable tungsten carbide, or a suitable stainless steel hardened to a minimum of 50 Rockwell "C" and precision ground and polished to a fine finish to prevent leakage and wear.

3.8.2.3 Pump capacity - The pump shall be capable of producing hydraulic pressure capable of spraying all types of coating materials with the exception of texture and fibrous paints, with a paint flow of 50 to 60 gph, using the necessary air flow at not more than 125 psi. The hydraulic pressure shall be constant and shall not fluctuate more than $\pm$ 5% at the spray gun nozzle.

3.8.2.4 Air filter and oiler - The pump shall be equipped with an air filter and an oiler. The minimum capacity of the air filter shall be not less than 25 cfm air flow at 100 psi. The oiler shall have a reservoir of sufficient size to hold a quantity of oil for not less than 8 hours of operation at maximum pump capacity. The oiler shall evenly dispense and distribute the oil so as to prevent binding or sticking of the pump components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.8.2.5 Pump cylinders - The pump cylinders shall be made of chromium plated high quality steel, nitralloy, or a material of an equivalent wear and abrasion resistance. The cylinder walls shall have a surface finish of not more than 15 rhr (roughness height rating).

3.8.2.6 Pump pistons - The pump pistons shall be made of high alloy steel hard chrome plated, or a material of an equivalent wear and abrasion resistance. Metallic surface finish shall be not more than 10 rhr and shall have a surface hardness of approximately 1050 Knoop or 1310 Vickers.

3.8.2.7 Pump air pressure regulator - The pump shall be equipped with one or more air pressure regulators of sufficient size to accommodate air flow from the minimum to the maximum capacities required for the pump.

3.8.2.7.1 Power stroke regulator - The power stroke regulator shall have a capacity of not less than 20 cfm air flow at 80 psi and shall be capable of regulating air pressure from 0 psi to 125 psi. The regulator shall be equipped with a dial type pressure gage with a corrosion-resistant mechanism. The gage shall be not less than 2 inches in diameter and shall be accurate within ±5% over the operating range of 25 to 125 psi.

3.8.2.8 Cleaning - The pump shall be equipped with a means of pressure cleaning.

3.8.2.9 Fittings - The spray outfit shall be equipped with standard connections to allow the attachment of hoses for the spray gun and air supply. The connections shall be of sufficient size to allow the pump to operate at its maximum capacity (see 3.8.2.3).

3.8.2.10 Packing - The pump packing shall be of the "V" or chevron type, and shall not be adversely affected by the various coating materials, thinners, or solvents. The packings shall be adequately stacked and backed for effective wiping and sealing. The pump packing shall withstand the pressure to which it is subjected without deformation or leakage.

3.8.3 Paint filter - The spray outfit shall be equipped with a paint filter at the outlet to the pump. The filter shall be capable of filtering not less than 80 gph of paint at not less than 100 micron filtration. It shall be constructed to provide flow of the paint in a manner to prevent pigment fall-out and settlement. When necessary, the filter shall be equipped with a drain valve to allow flushing of the filter when changing paint or cleaning the spray outfit. The filter shall be constructed to provide for quick and easy removal of the filtering mechanism. All filter components shall

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the current version before use.

be precision machined to provide for metal-to-metal or teflon seals. The filter shall be capable of operation at the maximum pressure developed by the pump and shall be constructed to provide ample support for all components to prevent collapse.

3.8.4 Spray gun - Each spray outfit shall be provided with one spray gun. The spray gun shall be of one piece construction forged from heat treated aluminum alloy or high tensile bronze. The gun shall weigh approximately 2 pounds, and shall have a trigger pull of less than 1/4 inch at a force of approximately 5 pounds. The gun shall be designed for primary operation in the range of 1000 to 2000 psi, and shall be capable of operating at pressures up to 2500 psi. The gun shall withstand a hydrostatic test of 10,000 psi without damage or distortion. The gun shall be capable of operation at the rate of 4 to 24 gph, using different size nozzles. Spray painting shall be accurately controlled with any given nozzle at a given pressure. Material passages shall be adequate for material demand required for satisfactory spraying with a minimum drop in pressure. The spray gun shall satisfactorily spray all spray materials on vertical, horizontal, overhead, or intermediate surfaces, without abnormal fatigue to the operator. It shall be sufficiently rugged to adequately withstand rough usage while supporting all other parts of the gun. All working parts shall be replaceable without the use of special tools. Threads of parts subject to wear shall be of strength and abrasion resistance not less than corrosion resisting steel. The gun shall be in the shape of a pistol grip suitable for full-sized two or more finger control. The spray fluid may enter the gun near the tip or at the bottom of the hand grip. A hook shall be provided at the top for hanging the gun when not in use. The gun shall consist essentially of a body, head, connector, nozzle assembly, material valve, trigger, and safety device.

3.8.4.1 Connector - The gun shall be provided with a 1/4-18 NPSM (M) male connector. A straight 360-degree swivel connection shall be provided for use between the material hose and the gun. Unless otherwise specified, all connectors shall be pressure fittings with 60-degree bevel seats. When specified (see 6.2), 30-degree bevel seats shall be provided. The connection fittings shall be made of corrosion-resistant steel or other suitable heavy-plated material which will withstand the high pressure and high abrasive paint action and shall be forged or machined from bar stock. There shall be no filters in the swivel.

3.8.4.2 Nozzle assembly - The nozzle assembly shall consist essentially of a coating material needle valve, a spray tip, and necessary auxiliary parts (see 6.2) suitable for spraying the materials specified. These parts, with the exception of the needle valve, shall be permanently and legibly marked with the manufacturer's size designation. The spray tip provided shall be designed for the type of spray material to be used and for specified spray pattern. Material to be sprayed shall be designated by the procuring activity (see 6.2). The spray gun nozzle assembly shall be constructed of tungsten carbide and the orifices shall be precision bored to control the spray patterns. The spray outfit shall be furnished with three to six nozzles covering a fan width range in increments of approximately 7 to 13 inches measured 8 inches from the nozzle tip.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the currentversion before use.

3.8.4.2.1 Coating material valve - The coating material valve shall be operated by the trigger and shall be held normally closed by coil springs enclosed within the gun body. Valves shall seat to cut off liquid flow. The valve stem shall be made of corrosion-resistant steel or equal and of a diameter suitable for the purpose intended. The valve stem and fluid nozzle orifice shall be in positive alignment so as to insure proper seating of the tip in order to prevent leakage when the trigger is released. The valve, seat and other parts subject to abnormal wear, shall be tungsten carbide or other material with equal or superior wear resistant characteristics and strength.

3.8.4.2.1.1 Valve packing - Valve packing shall be teflon or graphited type or any suitable material which will not be adversely affected by lacquer, vinyls, solvents, or paint fluids.

3.8.4.2.1.2 Needle assembly - The needle assembly shall be made of materials which will not be adversely affected by spray fluids. The needle tip or seating end shall be made of a wear-resistant material equal to tungsten carbide. The fluid needle shall properly seat to seal off the flow of spray fluids when the trigger is released by the operator. The fluid needle shall be easily and readily replaceable without the use of special tools.

3.8.4.2.2 Spray tips - The spray tips shall be made of tungsten carbide or of stainless steel or other metal with a tungsten carbide insert. The spray tip and orifice shall be ground or lapped on all seating surfaces so as to insure positive shut off without any leakage of spray fluids when the trigger is released. All interior surfaces affecting the material flow shall be smooth and finished to the manufacturer's standard tolerance. Spray tips shall be easily removable from the nozzle without the use of special tools. Spray tips shall provide a uniform spray pattern when the gun is held in the vertical, horizontal or other normal working positions.

3.8.4.2.3 Auxiliary parts - Baffle plates, tension washers, retainer rings, and other auxiliary parts shall be of non-corrosive material suitable for the purpose intended.

3.8.4.3 Trigger - The trigger shall be designed for not less than two finger control and shall operate to hold the fingers in normal position when spraying, and to aid in balancing the gun. Trigger control shall operate free and positive. Surface areas in contact with the fluid valve shall be hardened to resist wear. The trigger shall be reinforced, made of steel, heavy chrome plated or of a strong casting of manganese bronze alloy or heat treated aluminum alloy, polished to a bright finish.

3.8.4.4 Spray gun safety device - The gun shall be equipped with a safety device to prevent the flow of an injurious high pressure stream of spray fluid in the event the gun is accidentally triggered.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the current version before use.

3.8.5 Hose - The spray outfit shall be provided with one spray gun hose assembly. The spray gun hose assembly shall consist of one 25-foot long, 1/4 inch I.D. teflon hose. Each teflon hose shall be reinforced with stainless steel braiding or stainless steel covered flexible wire braiding. A static ground wire shall constitute a part of the braid. The hose assembly shall be capable of withstanding a burst pressure of 10,000 psi and a working pressure of 2500 psi. The hose assembly shall be equipped with fittings suitable for connection to the spray gun and pump units. Connections shall be permanently attached to each end of the hose with the static wire attached to the connections. The hose connections shall be 1/4-18 NPSM (F) female swivel, standard straight pipe thread with a 60-degree bevel seat to insure a tight, leakproof compression fit. Connections shall be made of corrosion-resistant steel or other ferritic alloys which are adequately zinc, cadmium, or chrome plated to resist corrosion, and shall be forged or turned from bar stock. External clamps on the connections are not acceptable.

3.8.6 Carriage assembly - The spray outfit shall be mounted on a carriage assembly having a metal base, frame, or platform of sufficient size and strength to support the spray outfit. The carriage assembly shall be designed for maximum access to the components mounted thereon and shall be equipped with a handle to facilitate steering. Two rubber tired wheels of at least 15 inches in diameter and one ball bearing swivel type rubber or plastic caster shall be provided to give maximum mobility and ease of operation. The wheels shall be equipped with a pedal-operated brake which will lock both wheels to prevent the spray outfit from moving on a 20 degree sloped surface.

3.8.7 Tool box - A tool box shall be provided as an integral part of each spray outfit. The tool box shall be of sufficient size to conveniently contain all tools required for the maintenance of the spray outfit, paint spray nozzles specified in 3.8.4.2 and standard 8-1/2 by 11 inch technical manuals. The tool box shall be manufactured of sheet metal of sufficient gauge to withstand rough handling. A hinged lid with a hasp shall be provided.

3.9 Part numbering of interchangeable parts - All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000/1 shall govern the manufacturer's part numbers and changes thereto.

3.10 Dimensions - The spray outfit shall not be more than 30 inches wide, 26 inches deep and 45 inches high.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

3.11 Weight - The weight shall not exceed 120 pounds.

3.12 Finish - All surfaces shall be cleaned of all grease, oil, scale, rust, dirt, casting sand, slag, and other extraneous matter and shall be dry. External surfaces shall have a smooth finish free from burrs, flaws, ragged and sharp edges.

3.12.1 Protective finish - The finish shall provide adequate protection against corrosion or other damage by weather elements. Parts made from ferrous material, unless otherwise specified, shall be made corrosion resistant with cadmium, chromium, or nickel plating.

3.12.2 Surface finish - Functional surfaces shall be smooth and free from burrs, or other harmful extraneous materials. Castings, forgings, molded, or welded parts, and all other metal parts shall be thoroughly cleaned and free from sand, dirt, fins, sprues, scale flux, or other harmful materials.

3.13 Identification of product -

3.13.1 Identification plate - A corrosion-resistant metal nameplate shall be permanently attached to the cleaning machine. The identification marking shall be in accordance with MIL-STD-130, and contain the following information.

Spray Outfit, Paint, Airless, Hydraulic Pressure, Portable

Contract or Order No.

Manufacturer's Name or Trademark

U.S.

3.14 Workmanship - Workmanship shall be of the quality necessary to produce complete units of good appearance, with safe and efficient operating characteristics as is customarily found in high grade commercial quality items of a similar nature.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection - The examination and testing of the spray outfit shall be classified as follows:

(a) First article inspection (4.3).

(b) Quality conformance inspection (4.4).

4.3 First article inspection - First article inspection shall consist of all the inspections of this specification.

4.3.1 Sampling for first article inspection - Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish one spray outfit for inspection to determine conformance with this specification. The sample shall be identified with the manufacturer's part number and contract number and shall be inspected as specified herein and in accordance with the contract or order (see 6.2).

4.4 Quality conformance - Each spray outfit on the contract or order shall be subjected to examination (4.6.1) and the spray tests (4.6.3). Any spray outfit containing a defect or failing to pass any of the inspections shall be rejected.

4.5 Test conditions - Unless otherwise specified, tests shall be conducted at an ambient temperature of 77 $\pm$ 18°F and at an ambient barometric pressure of 28 to 32 inches of mercury.

4.6 Inspection methods -

4.6.1 Examination - The spray outfit shall be thoroughly examined to determine conformance with the requirements of this specification not covered by tests specified herein.

4.6.1.1 The spray outfit shall be examined to ascertain that the preparation for delivery conforms to this specification.

4.6.2 Coating test - The spray outfit shall be operated using enamel, lacquer, oil base paint, and vinyl base paint. The spray outfit shall satisfactorily spray these various materials.

4.6.3 Spray tests - For the following tests, the spray outfit shall be charged with enamel paint conforming to TT-E-529. The spray outfit shall be connected to a source of compressed air of 125 psi at 20 cfm. The air pressure shall be adjusted to create proper atomization of material being sprayed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

4.6.3.1 Spray pattern test - The spray outfit shall be operated with each of the various spray nozzles supplied with it. The patterns of each nozzle shall be a spray fan ranging from 7 to 13 inches when measured 8 inches from the nozzle tip.

4.6.3.2 Paint flow capacity test - The outfit shall be operated and the paint flow shall be measured. The paint flow shall not be less than 50 or more than 60 gph.

4.6.3.3 Coating application test - The spray outfit shall be operated with each of the various spray nozzles supplied with it, and shall be used to coat flat and irregular metallic surfaces. The amount of coating material deposited with two passes of the spray gun, 12 inches from the surface, shall constitute a one coat application. There shall be no evidence of runs, sags, drips, or uneven coverage after a one coat application.

4.6.4 Leakage tests -

4.6.4.1 Test on gun - Leakage tests shall be performed by subjecting the spray outfit to actual operating conditions. The paint container shall be filled to capacity with enamel and the pump set in operation. After reaching working pressure, the gun shall be checked for leakage at all connections and at the needle valve locations. The gun shall be subjected to working pressures for a period of 10 minutes. Any seepage in excess of 3 drops per 10 minute period shall be cause for rejection. The trigger of the gun shall be actuated 3 times for 1/2 minute cycles to insure correct seating performance of the valves.

4.6.4.1.1 With the trigger safety device engaged, an attempt to operate the gun shall be made. There shall be no leakage or spray from the gun.

4.6.4.2 Test of spray outfit - Each connection of the unit shall be similarly tested by placing the unit under actual operating conditions. Leakage at piston rod packing shall not exceed 3 drops per minute. There shall be no evidence of leakage at any connection.

4.6.4.3 Inspection port - The closeable inspection port shall be checked. There shall be no evidence of leakage.

4.6.5 Hose test - The spray gun hoses shall be pressurized with water or a suitable hydraulic oil to a pressure of 5000 psi. At this pressure the hoses shall be subjected to a fluctuating pressure of 500 psi, for a minimum of 1000 cycles at the rate of not less than 10 cycles per minute. There shall be no evidence of leakage or visible signs of failure on the hose.

4.6.6 Continuous operation test - Standard atmospheric pressure and a temperature of 70° to 80°F shall prevail during the test specified herein. The spray

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

outfit shall be operated for 5 consecutive days in the following manner: 8 to 9 hours continuous operation at maximum capacity and shut down for 15 to 16 hours. The outfit shall pump an enamel paint with the gun so fixed as to spray the material back into the paint reservoir. The outfit, including the gun and hoses, shall not be drained until the 5 day test is completed. Upon completion of this test, the outfit shall be disassembled for inspection. There shall be no evidence of damage, defects, failure or excessive wear.

4.6.7 Load test - The paint container shall be loaded to capacity with paint, and then lifted by its handles. There shall be no deflection or permanent set in the container.

4.6.8 Carriage assembly brake test - The spray outfit, with the 10 gallon paint container filled to capacity, shall be placed in a forward position on a surface having at least a 20° slope and the brakes shall be applied. There shall be no movement of the carriage assembly. The test shall be repeated with the position of the spray outfit reversed.

5. PREPARATION FOR DELIVERY

5.1 Cleaning, preservation, and packaging - All spray outfits shall be cleaned in accordance with procedure C-1, dried in accordance with procedure D-1 or D-2 and packaged in accordance with method III of MIL-P-116. Preservation shall be provided in accordance with MIL-P-116 with the exceptions as noted herein.

5.1.1 Spray gun - The spray gun will not require preservation. The gun shall be packaged in accordance with method III of MIL-P-116.

5.1.2 Hose - Ends of hose shall be sealed with pressure sensitive tape conforming to PPP-T-60. Hose shall be packaged in accordance with method III of MIL-P-116.

5.1.3 Pump - The pump shall have all unpainted surfaces subject to corrosion given an application of preservative conforming to type P-2 of MIL-P-116. Regulators, valves, gages which may be damaged while on the pump or while in transit shall be removed. The pump shall be packaged in accordance with method III of MIL-P-116.

5.1.4 Fittings, regulators, valves, and gages - All fittings, regulators, valves, and gages having surfaces subject to corrosion shall be treated with P-9 preservative and packaged in accordance with method 1A of MIL-P-116. Openings shall be sealed with pressure sensitive tape conforming to type III, class I of PPP-T-60.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

When not assembled on the cover assembly or pump, fittings, gages, regulators, and valves shall be packaged in separate containers, and packaged in the same container with the cover assembly or pump.

5.1.5 Carriage assembly - Unpainted surfaces of the carriage assembly which are subject to corrosion shall be given an application of P-2 preservative in accordance with MIL-P-116. The carriage assembly shall be packaged in accordance with method III of MIL-P-116. Wheels shall be removed from the carriage assembly and packaged in such a manner that they will not be subjected to the weight of other components.

5.2 Packing -

5.2.1 Level A - All components comprising one complete spray outfit shall be packed in a box conforming to PPP-B-601 (overseas type), or PPP-B-621 (Class 2). Blocking, bracing, and anchoring shall be in accordance with MIL-STD-1186. Boxes shall be closed and strapped in accordance with the applicable box specification or the appendix thereto.

5.2.2 Level B - All components comprising one complete spray outfit shall be packed in a shipping container in such a manner that will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at the lowest rate.

5.3 Marking - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use - The spray outfits are intended to be used for painting aircraft ground support equipment.

6.2 Information for the contracting officer - Contracts or orders should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable level of cleaning, preservation, packaging and packing required (see 5.1, 5.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

(c) Special marking, if required (see 5.3).

(d) Items of data required (see 6.3).

(e) Hose connection bevel taper seat if other than specified (see 3.8.4.1).

(f) Auxiliary parts to nozzle assembly when required by spray pattern (see 3.8.4.2.3).

(g) Type material to be sprayed (see 3.8.4.2).

(h) Pump screen, size and quantity (see 3.8.2).

(i) When extra nozzles or tools are required (see 3.8.4.2).

(j) Engineering change procedure in accordance with NAVAIR INST 4275.3A. The Naval Air Systems Command, Dept. of the Navy, Washington D.C. 20360 (attn: AIR-534), will be responsible for review of class I changes under the contract. The Commanding Officer, Naval Air Engineering Center, Philadelphia, Pa. 19112 (attn: GSED) will be responsible for approval of class II changes.

6.2.1 Contracts or orders should specify the following provisions for first article inspection.

6.2.1.1 Whether first article inspection is required. When a contractor is in continuous production of the spray outfit from contract to contract, consideration should be given to waive the first article inspections. If first article inspection is required, indicate:

(a) Where the first article inspection is to be conducted (at the contractor's plant or Government or commercial laboratory).

(b) That the approval of first article samples or the waiving of the first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract.

6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification, (c) or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the current version before use.

(Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVAIR Form 4200/25 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Instruction manual - The instruction manual shall completely describe the operation of the spray outfit and include a description of the operation of each individual system. The instruction manual shall completely describe each maintenance operation.

6.4 Supersession data - This specification includes the requirements of purchase description XWS-4948, dated 6 April 1964.

Preparing Activity
Navy - AS
(Project No. 4940-N243)

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:37Z
Check the source to verify that this is the currentversion before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-81749(AS) SPRAY OUTFIT, PAINT, AIRLESS, HYDRAULIC PRESSURE, PORTABLE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8878

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Air Systems Command
Washington, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL AIR SYSTEMS COMMAND (AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:37Z
Check the source to verify that this is the current version before use.